ST12

Environment, Ecology, EIA, Biodiversity and Sustainable Development

Science and Technology is 18-23 questions in every paper and environment alone supplies 4-6 of them. Documented repeats: alpha/beta/gamma diversity (2013, 2016, 2018), greenhouse gases ranked by contribution (2016 and 2018), the acid rain pollutant pair (2016 and 2018 - the identical question twice), Ramsar sites of Rajasthan (2016, 2023), tiger reserves of Rajasthan (2023, 2024).

This is the biggest chapter in Science and Technology and the one RPSC mines hardest. Read it as three stacked layers - first the ecology (how an ecosystem actually runs), then the law (what India has enacted to protect it), then the diplomacy (what the world has agreed on climate). Almost every question is a definition, a year, a rank order or a matching pair, so the marks are there for anyone who memorises cleanly.

Environment & Ecology

Ecosystem
- Tansley 1935
- Grazing vs detritus chain
- 10% law - Lindeman 1942
- Energy pyramid always upright
- Gaseous vs sedimentary cycles

Biodiversity
- Genetic / species / ecosystem
- Alpha - beta - gamma (Whittaker)
- Hotspots - Myers 1988
- India's four hotspots
- IUCN 9 categories; threatened = CR+EN+VU

Conservation
- In-situ vs ex-situ
- Core - buffer - transition
- Project Tiger 1973, NTCA 2006
- Rajasthan - 5 tiger reserves
- Rajasthan - 5 Ramsar sites (2025)

Environmental law
- WPA 1972 - 4 schedules after 2022
- Water 1974, Air 1981
- EPA 1986 - umbrella, Art. 253
- BD Act 2002 + 2023 amendment
- FCA 1980 renamed in 2023
- NGT Act 2010

Pollution
- AQI 2014 - 6 bands, 8 pollutants
- NAAQS 2009, NCAP Jan 2019
- Acid rain = SO_2 + NO_x
- BOD 5 days at 20 deg C
- SUP ban 1 July 2022

Climate change
- CO_2 > CH_4 > CFC > N_2O
- SF_6 highest GWP
- IPCC 1988, UNFCCC 1992
- Kyoto 1997, Paris 2015
- COP30 Belem, COP31 Antalya
- Net-zero 2070; NAPCC 8 missions

EIA & disasters
- Notification 2006 under EPA
- Screening-Scoping-Consultation-Appraisal
- Category A central, B state
- Public hearing by SPCB
- DM Act 2005, Sendai 2015-30

The ecosystem – structure, energy flow and the 10 per cent law

Start with the word itself. An ecosystem is a community of living organisms plus the non-living surroundings they interact with, working as one functional unit. The term was coined by A.G. Tansley in 1935 - that name alone is a one-mark question. Around it runs a ladder of organisation that RPSC asks you to order: individual, population, community, ecosystem, landscape, biome, biosphere. Learn it upward, from the smallest to the largest.

- Structure of an ecosystem = biotic components (producers, consumers, decomposers) + abiotic components (light, temperature, water, soil, nutrients).
- Functions of an ecosystem = productivity, decomposition, energy flow, nutrient cycling. Four functions, learn all four.
- Producers are autotrophs; consumers are heterotrophs; decomposers such as bacteria and fungi are saprotrophs.
- Gross Primary Productivity minus respiration = Net Primary Productivity. Only NPP is available to the herbivore.
- Energy flow is one-way (unidirectional); nutrients go round in cycles. Never mix the two.

Now the food chain. Two types, and the difference is only in where the chain begins. The grazing food chain starts with a green plant. The detritus food chain starts with dead organic matter. Here is the point students get wrong - in a terrestrial ecosystem far more energy flows through the DETRITUS chain, because most leaves are never eaten alive; they fall and rot. In many aquatic systems the grazing chain dominates. Energy transfer between successive levels then obeys the 10 per cent law, given by Raymond Lindeman in 1942: only about a tenth of the energy at one trophic level reaches the next, the rest being lost as heat in respiration. That is exactly why a food chain rarely runs beyond four or five links.

- Trophic level 1 producers, level 2 herbivores (primary consumers), level 3 secondary consumers, level 4 tertiary consumers.
- 10 per cent law - Raymond Lindeman, 1942. The ecological pyramid concept - Charles Elton.

- Pyramid of energy is ALWAYS upright, in every ecosystem, because of the 10 per cent law.
- Pyramid of numbers can be inverted - one big tree supports many insects and still more parasites.
- Pyramid of biomass is inverted in the open ocean - a small standing crop of phytoplankton supports a larger mass of zooplankton and fish.

TRAP

Two traps live in this one page. First: it is the pyramid of ENERGY that is always upright, not biomass and not numbers. Second: Lindeman gave the 10 per cent law, Elton gave the ecological pyramid, Tansley coined 'ecosystem', Odum wrote the textbook. RPSC has offered all four names in a single option set.

Nutrient cycles, succession and the special species categories

Nutrients move in closed loops called biogeochemical cycles. Split them into two families by where the main reservoir sits. If the reservoir is the atmosphere it is a gaseous cycle; if it is rock and sediment it is a sedimentary cycle. The phosphorus cycle is the perennial favourite because it has no significant gaseous phase at all.

Gaseous cycle vs sedimentary cycle - the standard question

Gaseous cycle	Sedimentary cycle
Reservoir is the atmosphere	Reservoir is the earth's crust - rocks and sediments
Carbon, nitrogen, oxygen	Phosphorus, sulphur
Fast, well buffered, global in reach	Slow, easily disturbed, local in reach
Has a gaseous phase	Phosphorus has NO gaseous phase

Nitrogen fixers matched to their mode – a standing matching set

Organism	Mode of nitrogen fixation
Rhizobium	Symbiotic, in the root nodules of legumes
Frankia	Symbiotic, with Casuarina
Azotobacter	Free-living, aerobic bacterium
Clostridium	Free-living, anaerobic bacterium
Anabaena / Nostoc	Cyanobacteria; Anabaena azollae lives in the water fern Azolla

- Ecological succession – the orderly replacement of one community by another until a stable climax community is reached.
- Primary succession begins on a lifeless surface; secondary succession begins where soil survives, and is much faster.
- Xerarch succession starts on bare rock and its pioneers are LICHENS; hydrarch succession starts in water and its pioneers are phytoplankton.
- Ecotone = the transition zone where two ecosystems meet, such as a mangrove between land and sea.
- Edge effect = the higher species density found in an ecotone; species confined to it are edge species.
- Keystone species – an influence far out of proportion to its numbers (sea otter, fig trees).
- Flagship species – charismatic, used to raise public support (tiger, Great Indian Bustard).
- Umbrella species – needs so much area that protecting it protects a whole community (tiger, elephant).
- Indicator species – its condition reveals environmental quality; lichens indicate clean air.
- Invasive alien species in India – Lantana camara, Parthenium (congress grass), Eichhornia or water hyacinth ('terror of Bengal'), Prosopis juliflora.

Biodiversity - levels, the alpha-beta-gamma trio and the hotspots

Biodiversity means the variety of life, measured at three levels. Genetic diversity is variation within a species - the thousands of rice landraces. Species diversity is the number and abundance of species in an area. Ecosystem diversity is the variety of habitats in a region. The word itself was coined by Walter Rosen in 1985 and popularised by E.O. Wilson. On top of that sits R.H. Whittaker's alpha-beta-gamma trio, and this is a documented RPSC repeat. It is a scale story, not a difficulty story. Alpha is diversity INSIDE one habitat. Beta is the CHANGE in species as you move from one habitat to the next along a gradient. Gamma is the TOTAL for the whole landscape. Because gamma is the regional total, it can never be smaller than the alpha of a single habitat within it - RPSC has built a deliberately wrong statement on exactly that point.

Whittaker's three measures of diversity

Measure	What it counts	Scale
Alpha diversity	Species within a single community or habitat	Local, within-habitat
Beta diversity	Turnover in species composition between communities along a gradient	Between habitats
Gamma diversity	Total diversity of a large landscape or geographical region	Regional, the sum total

ASKED IN RAS

Asked in RAS Pre 2018 - the diversity of species found within a single community or habitat is termed? Answer: alpha diversity. Asked again in RAS Pre 2016 as a statement question, where 'alpha diversity of a habitat is always greater than the gamma diversity of its region' was the wrong statement. Add the 2013 hotspot question and this cluster has appeared in three papers - treat it as certain.

A biodiversity hotspot is not simply a rich area; it is a rich area in danger. Norman Myers proposed the idea in 1988 and there are two criteria, both of which must be met together. About 36 hotspots are recognised worldwide and four of them reach into Indian territory.

- Criterion 1 - at least 1,500 species of endemic vascular plants. Criterion 2 - at least 70 per cent of its primary vegetation already lost. Both, together.
- India's four hotspots - the Himalaya, Indo-Burma, the Western Ghats and Sri Lanka, and Sundaland.
- Sundaland enters India ONLY through the Nicobar Islands, not through the mainland.
- Cerrado (Brazil), Caucasus, Madagascar, Mesoamerica and the Cape Floristic Region do NOT extend into India - RAS Pre 2013 asked exactly this, with Cerrado as the answer.
- Rajasthan lies in none of the four; the Aravalli is ecologically vital but is not a hotspot.
- Trick to remember India's four - 'HIWS': Himalaya, Indo-Burma, Western Ghats-Sri Lanka, Sundaland.

IUCN Red List and the two styles of conservation

The International Union for Conservation of Nature was founded in 1948 and works out of Gland in Switzerland. Its Red List places every assessed species in one of nine categories. You must be able to run the ladder in order, and you must know which three of them together make up the word 'threatened'.

IUCN Red List - the ladder from safe to gone

Least Concern (LC) → Near Threatened (NT) → Vulnerable (VU) → Endangered (EN) → Critically Endangered (CR) → Extinct in the Wild (EW) → Extinct (EX)

- Nine categories in all - the seven above plus Data Deficient and Not Evaluated, which sit outside the risk ladder.
- 'Threatened' = Critically Endangered + Endangered + Vulnerable. Exactly these three, no more, no fewer.
- Extinct in the Wild means it survives only in cultivation, captivity or outside its natural range.
- The Great Indian Bustard (Godawan), Rajasthan's state bird, is Critically Endangered - the single most examinable Red List entry for this state.
- The cheetah became extinct in India in 1952; African cheetahs were reintroduced at Kuno, Madhya Pradesh, from 2022.
- CITES (1973, in force 1975) regulates international trade in species through three appendices - do not confuse it with the IUCN Red List, which only assesses risk.

In-situ vs ex-situ conservation

In-situ (on site)	Ex-situ (off site)
Species protected in its natural habitat	Species maintained outside its natural habitat
National parks, sanctuaries, biosphere reserves, conservation and community reserves, sacred groves	Zoos, botanical gardens, seed and gene banks, cryopreservation of gametes, tissue culture, captive breeding
The whole ecosystem is conserved along with the species	Only the species or its germplasm is conserved
Cheaper, self-sustaining; evolution continues	Costly, but the only route for species already too rare in the wild

- Biosphere reserve zones - core (legally protected, no human activity), buffer (limited research and education), transition (settlement and sustainable use).
- India's first biosphere reserve - Nilgiri, notified in 1986. Eighteen have been notified in all and twelve are on UNESCO's World Network.
- Rajasthan has NO biosphere reserve. Learn this negative fact; it is exactly what RPSC asks.
- Nokrek - Meghalaya; Simlipal - Odisha; Dihang-Dibang - Arunachal Pradesh; Agasthyamalai - Kerala and Tamil Nadu.

Protected areas, species projects and Rajasthan's wetlands

India's species projects are simple year-matching questions. Project Tiger came first, in 1973, with nine reserves - and Ranthambore was one of those original nine, a Rajasthan link RPSC has used. The National Tiger Conservation Authority is not a separate creature: it was given statutory status in 2006 through an amendment to the Wild Life (Protection) Act, 1972. Wetlands are the other half of this section, and Rajasthan's answer here has changed recently, so read the current-affairs box carefully.

- Project Tiger 1973 (nine reserves at launch, including Ranthambore); NTCA statutory in 2006 under the WPA 1972.
- Project Crocodile 1975, Project Elephant 1992, Project Snow Leopard 2009, Project Dolphin 2020.
- The all-India tiger estimation of 2022 put the population at about 3,682.
- Protected area categories under the WPA - national park, wildlife sanctuary, conservation reserve, community reserve.
- Ramsar Convention on Wetlands - signed 1971 at Ramsar in Iran, in force from 1975; a site in trouble goes on the Montreux Record.

- Keoladeo Ghana is both a Ramsar site and a UNESCO World Heritage Site; Sambhar is India's largest inland saline lake.

Tiger reserves of Rajasthan - district and year

Tiger reserve	District	Note
Ranthambore	Sawai Madhopur	One of the original nine of 1973
Sariska	Alwar	Site of the world's first successful tiger relocation, 2008
Mukundra Hills	Kota	Notified 2013
Ramgarh Vishdhari	Bundi	Notified 2022, India's 52nd
Dholpur-Karauli	Dholpur and Karauli	Approved 2023, India's 54th; Rajasthan's fifth

ASKED IN RAS

Asked in RAS Pre 2024 as a matching question - Ranthambore with Sawai Madhopur, Sariska with Alwar, Mukundra Hills with Kota, Ramgarh Vishdhari with Bundi. And asked in RAS Pre 2016 and again in 2023 - which pair were Rajasthan's FIRST TWO Ramsar sites? Answer: Keoladeo and Sambhar. Learn the district against every reserve, and learn 'first two' as a phrase, because the total has since changed.

CURRENT LINK

CURRENT LINK, as of September 2026 - Rajasthan now has FIVE Ramsar sites: Keoladeo Ghana National Park (1981), Sambhar Lake (1990), and three added in 2025 - Menar in Udaipur, Khichan in Phalodi and Siliserh Lake in Alwar. India's national tally crossed a hundred and stood at 101 in August 2026. Two more datelines worth carrying into the hall: the India State of Forest Report 2023 (released December 2024) is still the latest assessment - forest and tree cover 8,27,357 sq km or 25.17 per cent of the geographical area, forest cover alone 21.76 per cent, mangroves 4,992 sq km, Madhya Pradesh largest by area and Lakshadweep highest by percentage; and Madhav in Madhya Pradesh became India's newest tiger reserve in 2025.

India's environmental laws - the table you must be able to write from memory

This section is pure marks. Every year RPSC asks either an Act-to-year match or one clause of a recent amendment. Learn the table first, then the three amendments that have changed the standard answers since 2022.

Environmental legislation of India - Act and year

Act	Year	The point that gets asked
Wild Life (Protection) Act	1972	The 2022 amendment cut the schedules from six to four, in force 1 April 2023
Water (Prevention and Control of Pollution) Act	1974	Created the CPCB and the State Pollution Control Boards
Forest (Conservation) Act	1980	Renamed by the 2023 amendment
Air (Prevention and Control of Pollution) Act	1981	Enacted to give effect to the Stockholm decisions

Act	Year	The point that gets asked
Environment (Protection) Act	1986	The umbrella Act; passed under Article 253 after Stockholm 1972 and the 1984 Bhopal disaster
Biological Diversity Act	2002	Gives effect to the Convention on Biological Diversity, 1992; amended 2023
Forest Rights Act (Scheduled Tribes and Other Traditional Forest Dwellers)	2006	Ministry of Tribal Affairs is nodal; Gram Sabha is the authority; cut-off 13 December 2005
National Green Tribunal Act	2010	NGT set up on 18 October 2010
Compensatory Afforestation Fund Act	2016	The CAF Fund has its OWN Act - not the 2023 forest amendment

- Wild Life (Protection) Amendment Act 2022 - six schedules cut to four. Schedules I and II are animals, Schedule III is plants, Schedule IV lists CITES specimens.
- Biological Diversity (Amendment) Act 2023 - exempts registered AYUSH practitioners from benefit sharing for codified traditional knowledge, and decriminalises offences, replacing imprisonment with monetary penalties.
- It did NOT abolish Biodiversity Management Committees; they continue and still maintain People's Biodiversity Registers.
- Forest (Conservation) Amendment Act 2023 - renamed the parent Act as the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980.
- Its two headline exemptions - forest land within 100 km of an international border for strategic linear projects of national importance, and up to 10 hectares for security-related infrastructure. Zoos, safaris and eco-tourism facilities were exempted too.

- Biological Diversity Act structure is three-tier: National Biodiversity Authority at Chennai, State Biodiversity Boards, Biodiversity Management Committees at local body level.

- NGT - principal bench New Delhi; zonal benches Bhopal, Pune, Kolkata and Chennai; cases to be disposed of within six months.
- India was the third country in the world to set up a dedicated environmental court, after Australia and New Zealand.
- The Environment (Protection) Act 1986 is the parent statute for almost every environmental rule you will meet - EIA, plastic, e-waste, noise.
- Environment Audit Rules were notified on 29 August 2025 under the same 1986 Act.

> **TRAP**
> Do not mix up the two amendments of 2023. The BIOLOGICAL Diversity amendment is about AYUSH practitioners and decriminalisation. The FOREST Conservation amendment is about the 100 km border exemption and the Hindi renaming. And the Compensatory Afforestation Fund has its own statute of 2016 - RPSC has floated 'the 2023 amendment gave statutory backing to the CAF' as a deliberately wrong statement.

Pollution - air, water, noise, plastic and e-waste

Air pollution questions come in three shapes: the AQI band table, the list of pollutants inside the index, and the acid rain pair. Take the AQI first. It was launched in 2014, is computed by the CPCB, has six categories and eight pollutants, and the AQI you see reported is the WORST sub-index among those eight - not their average.

National Air Quality Index - the six bands

AQI value	Category	AQI value	Category
0-50	Good	201-300	Poor
51-100	Satisfactory	301-400	Very Poor

AQI value	Category	AQI value	Category
101-200	Moderate	401-500	Severe

- The eight AQI pollutants - PM10, PM2.5, NO2, SO2, CO, ground-level ozone (O3), NH3 and Pb.
- Methane and carbon dioxide are NOT in the AQI. This is a favourite negative-framed question.
- National Ambient Air Quality Standards were notified in 2009 and cover twelve pollutants.
- National Clean Air Programme - launched January 2019 by the MoEFCC through the CPCB for identified non-attainment cities.
- NCAP's revised target - a 40 per cent cut in particulate matter concentration by 2025-26 over the 2017 level; the original was 20-30 per cent by 2024.
- Acid rain forms when SO2 and NOx convert in the atmosphere to sulphuric and nitric acid; rain with a pH below 5.6 counts as acidic.
- Stubble burning in Punjab and Haryana is the seasonal driver of the north Indian winter smog; indoor pollution from biomass chullhas is tackled through the LPG route.

ASKED IN RAS

Asked in RAS Pre 2016 and again word for word in RAS Pre 2018 - acid rain is caused mainly by which pair of pollutants? Answer: sulphur dioxide and oxides of nitrogen. Two consecutive papers, the identical question. If you learn one pollution fact, learn this one.

Water pollution turns on two abbreviations. BOD is the oxygen micro-organisms need to break down the organic matter in a sample, measured over five days at 20 degrees C; a high BOD means dirty water. COD measures everything oxidisable, biodegradable or not, so for the same sample COD is always higher than BOD. Eutrophication is the other standing question - nutrient enrichment feeds an algal bloom, and it is the DECOMPOSITION of that bloom that strips the oxygen out of the water and kills the fish. The bloom does not raise dissolved oxygen; students tick that wrong statement every year.

Pollutant matched to the disease it causes

Pollutant	Disease
Mercury	Minamata disease
Cadmium	Itai-itai disease
Fluoride	Skeletal fluorosis - a real problem in several western Rajasthan districts
Nitrate in drinking water	Blue baby syndrome (methaemoglobinaemia)
Arsenic	Black foot disease and skin lesions

- Eutrophication is driven chiefly by nitrates and phosphates from fertiliser runoff and sewage.
- Namami Gange - launched 2014 under the National Mission for Clean Ganga.
- Single-use plastic ban on identified items - effective 1 July 2022; carry bag thickness raised to 120 microns from 31 December 2022.
- EPR = Extended Producer Responsibility - the producer stays responsible for the product after the consumer has finished with it.
- E-Waste (Management) Rules 2022 came into force on 1 April 2023.
- Noise limits, day time, in dB(A) Leq - industrial 75, commercial 65, residential 55, silence zone 50; night values 70, 55, 45 and 40.
- Solid Waste Management Rules 2016 made source segregation into wet, dry and hazardous waste mandatory; hazardous waste has its own rules of 2016.
- Soil pollution comes from pesticide and fertiliser residue, industrial effluent and dumped waste; thermal pollution lowers dissolved oxygen; radioactive pollution is measured in becquerel and sievert.

Climate change - the gases, the treaties and India's promises

Two different rankings of greenhouse gases exist and RPSC has asked both, so keep them apart. The first is by CONTRIBUTION to the enhanced greenhouse effect - how much warming each gas is actually delivering, which depends on how much of it is up there. The second is by GLOBAL WARMING POTENTIAL - how strong one molecule is, regardless of quantity. The two orders run in roughly opposite directions.

- By contribution: carbon dioxide (about 60%) > methane (about 20%) > CFCs (about 14%) > nitrous oxide (about 6%).
- By 100-year global warming potential per molecule, sulphur hexafluoride (SF_6) is the highest of all.
- Water vapour is the most abundant greenhouse gas overall but is not counted among the anthropogenic gases whose contribution is ranked.
- The Kyoto basket - CO_2, CH_4, N_2O, HFCs, PFCs and SF_6; nitrogen trifluoride was added later.
- Methane comes largely from paddy fields, enteric fermentation in ruminants and landfills; nitrous oxide largely from nitrogenous fertiliser.
- The IPCC does not do research - it assesses published science. Its Sixth Assessment Report cycle (AR6) ran from 2021 to 2023.

ASKED IN RAS

Asked in RAS Pre 2018 - which gas makes the largest contribution to the enhanced greenhouse effect? Answer: carbon dioxide. Asked in RAS Pre 2016 as an ordering question - arrange methane, carbon dioxide, nitrous oxide and CFCs in descending order of contribution. Answer: CO_2, methane, CFCs, nitrous oxide. Same fact, two papers, two formats - so learn it both as a list and as a rank.

YAAD RAKHO

Mnemonic for the contribution order - 'Car Meets Cool Nights': Carbon dioxide, Methane, CFCs, Nitrous oxide. For potency, one word is enough - SF6 sits at the top of the GWP table.

The climate and environment chronology - with the one outcome each is asked for

Stockholm 1972 - first UN environment conference; UNEP follows, at Nairobi → Vienna Convention 1985 - framework for ozone → Montreal Protocol 1987 - phases out ozone-depleting substances → IPCC 1988 - set up by WMO and UNEP → UNFCCC, Rio 1992 - common but differentiated responsibilities → Kyoto Protocol 1997 - binding targets on Annex-I countries; CDM, JI, emissions trading → Paris Agreement 2015 - well below 2 deg C, pursue 1.5; voluntary NDCs, stocktake every 5 years → COP26 Glasgow 2021 - India's Panchamrit and net-zero 2070 → COP28 Dubai 2023 - first Global Stocktake; Loss and Damage Fund operationalised → COP29 Baku 2024 - climate finance goal of USD 300 bn a year by 2035 → COP30 Belem 2025 - the 'global mutirao' decision; no binding fossil-fuel roadmap in the text → COP31 Antalya, Turkiye 2026

NAPCC 2008 - the eight missions and their nodal ministries

Mission	Nodal ministry
National Solar Mission	New and Renewable Energy
National Mission for Enhanced Energy Efficiency	Power

Mission	Nodal ministry
National Mission on Sustainable Habitat	Housing and Urban Affairs
National Water Mission	Jal Shakti
National Mission for Sustaining the Himalayan Ecosystem	Science and Technology
National Mission for a Green India	Environment, Forest and Climate Change
National Mission for Sustainable Agriculture	Agriculture and Farmers Welfare
National Mission on Strategic Knowledge for Climate Change	Science and Technology

CURRENT LINK

CURRENT LINK, as of September 2026 - India's new Nationally Determined Contribution for 2031-35 was approved by the Union Cabinet on 25 March 2026: a 47 per cent cut in the emissions intensity of GDP by 2035 over the 2005 level, about 60 per cent of cumulative electric power capacity from non-fossil sources by 2035, and an additional carbon sink of 3.5 to 4.0 billion tonnes of CO2 equivalent. Net-zero stays at 2070. The older 2030 targets - 45 per cent intensity cut and 50 per cent non-fossil capacity - are still asked, so hold both sets. COP31 is scheduled at Antalya, Turkiye, in 2026. Every state also prepares a State Action Plan on Climate Change under the NAPCC umbrella, Rajasthan included. Remember also that there is NO 'National Clean Air Mission' inside the NAPCC's eight - NCAP is a separate 2019 programme, and that distractor has been used. India's other climate vehicles: the International Solar Alliance (2015, India and France, headquarters Gurugram - the first treaty-based international organisation headquartered in India), CDRI (2019, New Delhi), Mission LiFE (October 2022, Kevadia) and the Carbon Credit Trading Scheme notified in 2023 under the Energy Conservation (Amendment) Act, 2022.

Ozone, sustainable development and the SDGs

Keep the ozone story separate from the climate story - students merge them and lose the mark. Ozone depletion is chlorofluorocarbons and other halogenated compounds destroying stratospheric ozone; global warming is greenhouse gases trapping heat. Different problem, different treaty line. Sustainable development then has one definition that is asked almost verbatim: development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It comes from the Brundtland Commission report Our Common Future, 1987. Learn the report name and the year as one unit.

- Vienna Convention for the Protection of the Ozone Layer - 1985, the framework convention.
- Montreal Protocol - 1987, the operational protocol phasing out ozone-depleting substances. World Ozone Day is 16 September.
- Kigali Amendment - 2016, phases DOWN hydrofluorocarbons, which harm the climate but not ozone; in force 1 January 2019; India ratified in 2021.
- The ozone hole forms over Antarctica in the southern spring; ozone thickness is measured in Dobson units.
- The Montreal Protocol is the only environmental treaty with universal ratification - often called the most successful of them all.

- Sustainable Development Goals - 17 goals and 169 targets, adopted in September 2015 as part of the 2030 Agenda.
- They succeeded the eight Millennium Development Goals (2000-2015).
- SDG 13 is Climate Action, SDG 14 Life Below Water, SDG 15 Life on Land.
- In India the SDG India Index is published by NITI Aayog.
- Ecological footprint - the biologically productive area needed to support a person's consumption; when demand exceeds supply the region runs an ecological deficit.

- Green accounting adjusts national income for the depletion of natural capital and environmental damage.
- Convention on Biological Diversity - Rio 1992; the Nagoya Protocol (2010) covers access and benefit sharing; the Kunming-Montreal Global Biodiversity Framework was adopted in 2022.

Environmental Impact Assessment and disaster management

EIA is the process by which a project is examined for its environmental consequences BEFORE it is allowed to be built. In India it runs under the EIA Notification of 2006, issued under the Environment (Protection) Act, 1986. There are four stages in a fixed order, and RPSC's standard question is which stage does NOT belong.

The four stages of environmental clearance under the EIA Notification 2006

Screening → Scoping (Terms of Reference) → Public Consultation → Appraisal by the Expert Appraisal Committee

- Category A projects - appraised centrally by an Expert Appraisal Committee and cleared by the MoEFCC.
- Category B projects - appraised by a State Expert Appraisal Committee and cleared by the State Environment Impact Assessment Authority.
- Within Category B, B1 projects need a full EIA report; B2 projects do not.
- The public hearing is conducted by the State Pollution Control Board or Pollution Control Committee.
- 'Cost-benefit certification' and 'post-project audit' are NOT among the four stages - both have been used as distractors.

- The draft EIA Notification 2020 drew heavy criticism for allowing post-facto clearance and shortening the public notice period; applications are filed on the PARIVESH portal.
- Strategic EIA assesses a whole policy, plan or programme rather than one project; a carrying capacity study fixes the limit an area can absorb.

TRAP

Two clean traps here. First: the public hearing is conducted by the STATE POLLUTION CONTROL BOARD - not by the ministry, not by the SEIAA. Second: for Category B the SEAC only APPRAISES while the SEIAA GRANTS the clearance; RPSC swaps the two bodies and most students do not notice.

Bodies under the Disaster Management Act, 2005 - the head of government heads the authority

Body	Ex-officio head
National Disaster Management Authority (NDMA)	Prime Minister
State Disaster Management Authority (SDMA)	Chief Minister
District Disaster Management Authority (DDMA)	District Collector; the Zila Parishad chairperson is co-chairperson
National Executive Committee (NEC)	Union Home Secretary

- The DM Act 2005 also created the National Disaster Response Force and the National Institute of Disaster Management; NDMA has up to nine members besides the Prime Minister.
- India's first National Disaster Management Plan, aligned with the Sendai Framework, was released in 2016.
- Sendai Framework for Disaster Risk Reduction - adopted 2015, runs 2015-2030, has four priorities and seven global targets; it replaced the Hyogo

Framework (2005-2015).

- Disaster types - natural (earthquake, flood, cyclone, drought, landslide) and man-made (industrial, chemical, nuclear, stampede).
- Rajasthan's recurring hazards are drought, heat wave, flash flood in the Hadoti belt and lightning; the state runs an SDMA under the same Act.

REVISION RECAP

1. Ecosystem - term by A.G. Tansley, 1935. Ladder: individual, population, community, ecosystem, landscape, biome, biosphere.
2. 10 per cent law of energy transfer - Raymond Lindeman, 1942; the ecological pyramid concept - Charles Elton.
3. Pyramid of energy always upright; pyramids of number and biomass can be inverted.
4. Grazing chain starts with green plants, detritus chain with dead matter - and on land the detritus chain carries more energy; gaseous cycles are carbon, nitrogen and oxygen, sedimentary are phosphorus and sulphur, and phosphorus has no gaseous phase.
5. Rhizobium-legume, Frankia-Casuarina, Azotobacter free-living aerobic, Anabaena-Azolla.
6. Lichens pioneer bare rock; ecotone is the transition zone and the edge effect its extra richness.
7. Alpha within a habitat, beta turnover along a gradient, gamma the whole region - gamma is never less than alpha.
8. Hotspots - Myers 1988; 1,500 endemic vascular plants plus 70 per cent primary vegetation lost. India's four: Himalaya, Indo-Burma, Western Ghats-Sri Lanka, Sundaland (Nicobar).
9. IUCN 1948, Gland; nine Red List categories; threatened = Critically Endangered + Endangered + Vulnerable.
10. Biosphere reserve = core, buffer, transition; India's first Nilgiri 1986; Rajasthan has none.
11. Project Tiger 1973, NTCA statutory 2006; Rajasthan's five reserves - Ranthambore, Sariska, Mukundra Hills, Ramgarh Vishdhari, Dholpur-Karauli.
12. Rajasthan's five Ramsar sites - Keoladeo 1981, Sambhar 1990, and Menar, Khichan and Siliserh all in 2025 (as of September 2026).
13. Act years - WPA 1972 (four schedules after 2022), Water 1974, FCA 1980 (renamed 2023), Air 1981, EPA 1986, BD Act 2002 (amended 2023), FRA 2006, NGT Act 2010, CAF Act 2016.

14. AQI 2014 - six bands from Good 0-50 to Severe 401-500, eight pollutants; methane and CO2 are excluded.
15. Acid rain = SO2 + NOx, pH below 5.6; BOD over five days at 20 deg C; COD always exceeds BOD.
16. Greenhouse contribution order CO2 > CH4 > CFC > N2O; the highest GWP belongs to SF6.
17. Stockholm 1972, Vienna 1985, Montreal 1987, IPCC 1988, UNFCCC 1992, Kyoto 1997, Paris 2015, Belem COP30 2025, Antalya COP31 2026.
18. India - net-zero 2070; the 2035 NDC approved 25 March 2026 sets a 47 per cent intensity cut and 60 per cent non-fossil capacity; NAPCC 2008 has eight missions.
19. EIA 2006 - screening, scoping, public consultation, appraisal; public hearing by the SPCB; Category A central, B state, B1 needs an EIA report.
20. DM Act 2005 - NDMA under the PM, SDMA under the CM, DDMA under the Collector, NEC under the Union Home Secretary; Sendai 2015-2030 with four priorities and seven targets.